

1RV24MC042_HARSHAVARDHAN_BR_LAB7

Deploying a Web Application to Kubernetes Using Minikube

Folder Structure

```
nginx-k8s-deployment/  
  nginx-deployment.yaml
```

nginx-deployment.yaml

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          ports:  
            - containerPort: 80
```

Step 1: Start Minikube and Check Status

```
minikube start  
minikube status
```

```

> ls
nginx-deployment.yaml
> tree
.
├── nginx-deployment.yaml
1 directory, 1 file
> minikube start
minikube status
🐳 minikube v1.37.0 on Linuxmint 22.1
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...

🔥 Docker is nearly out of disk space, which may cause deployments to fail! (91% of capacity). You can pass '-
' to skip this check.
💡 Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🇺🇸 Related issue: https://github.com/kubernetes/minikube/issues/9024

🚧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, metrics-server, default-storageclass, dashboard
🏡 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubecfg: Configured

```

Step 2: Apply Configurations

```
kubectl apply -f nginx-deployment.yaml
```

Step 3: Verify Deployment and Pods

```
kubectl get deployments
kubectl get pods
```

```

> kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
> kubectl get deployments
kubectl get pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     0            0           7s
NAME                READY   STATUS    RESTARTS   AGE
program-8           1/1     Running    1 (30s ago)  11m

```

Step 4: Expose the Deployment as a Service

```
kubectl expose deployment nginx-deployment --type=NodePort --port=80
kubectl get services
```

```
> kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
> kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP        2m3s
nginx-deployment    NodePort    10.108.179.59 <none>       80:31176/TCP   13s
```

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```
> kubectl expose deployment nginx-deployment --type=NodePort --port=80
Error from server (AlreadyExists): services "nginx-deployment" already exists
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP        5m32s
nginx-deployment    NodePort    10.108.179.59 <none>       80:31176/TCP   3m42s
> minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31176

🌐 Opening service default/nginx-deployment in default browser...
Opening in existing browser session.

Step 5: Access the Service

```
minikube service nginx-deployment
```

```
curl http://localhost:<node-port>
```

```
> minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31176

🌐 Opening service default/nginx-deployment in default browser...
Opening in existing browser session.

```
> curl http://192.168.49.2:31176
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

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[p9] 0:zsh* "blaze" 10:59 19-Dec-25

Step 6: Clean Up Resources

```
kubectl delete pods --all
kubectl delete services --all
kubectl delete deployments --all
```

```
minikube stop
```

```
> kubectl delete pods --all
kubectl delete services --all
kubectl delete deployments --all
pod "program-8" deleted from default namespace
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
deployment.apps "nginx-deployment" deleted from default namespace
> minikube stop
🔥 Stopping node "minikube" ...
🔴 Powering off "minikube" via SSH ...
🔴 1 node stopped.
```